PRE-ALGEBRA (7TH) · UNIT 3: DECIMALS AND EQUATIONS

Road Trip Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan a road trip from Boston to Montreal. They estimate and compute with decimals, use d = rt, find means and medians of times, solve decimal equations for costs, and convert metric units.

Crossing into Canada gives a natural reason to convert kilometers, liters, and currency.

Standards Addressed

- 7.NS.A.3 Solve problems with rational numbers
- 7.EE.B.3 Multi-step problems with decimals; estimation
- 7.EE.B.4 One-step equations
- 7.RP.A.2 Rates and unit conversions

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a map of the route. Part 1 estimating and trip data.
Day 2	Part 2 formulas and money equations.
Day 3	Part 3 metric conversions and the itinerary.

Materials

- Student handout and Trip Leg Cards
- Calculators
- Road map or online map (optional)

How the Three Versions Differ

Level	What changes
CP2	Rounding, adding decimals, d = rt with one step, and simple conversions.
CP1	Leg times, mean and median, money equations, and metric and currency conversions.
Honors	Average speed vs. mean speed, fuel efficiency changes, tax equations, and multi-step metric rates.

Watch For (Common Misconceptions)

- Misplacing the decimal point when multiplying money.
- Writing 1.2 hours as 1 hour 20 minutes.
- Converting in the wrong direction (multiplying instead of dividing).

Extension Ideas

- Plan a trip to another city using real distances.
- Compare electric car charging costs.
- Graph distance vs. time for the trip.

Answer Key – CP2

Part 1: Estimate the Trip

#	Task	Answer
1	Round each US leg to the nearest mile. Estimate the total US distance.	68 + 70 + 95 = 233 miles
2	Find the exact total of the three US legs.	68.4 + 69.8 + 95.1 = 233.3 miles
3	Gas costs $3.39 per gallon. Estimate the cost of 10 gallons, then find it exactly.	About $34; exactly $33.90
4	Find the mean of the average speeds 57, 62, and 63.4 mph.	(57 + 62 + 63.4) ÷ 3 = 60.8 mph

Part 2: Distance, Rate, Time

#	Task	Answer
5	How far do you drive in 1.5 hours at 62 mph?	d = 62 × 1.5 = 93 miles
6	How long does the Boston → Concord leg take at 57 mph? Round to the nearest tenth of an hour.	t = 68.4 ÷ 57 = 1.2 hours
7	Solve 31.5g = 233.3 to find the gallons of gas for the US legs. Round to the nearest tenth.	g ≈ 7.4 gallons
8	How much does that gas cost at $3.39 per gallon?	7.4 × 3.39 ≈ $25.09

Part 3: Canada Stop

#	Task	Answer
9	Convert 153 km to miles.	153 × 0.62 ≈ 94.9 miles
10	Gas in Canada costs $1.62 per liter. About how much is that per gallon?	1.62 × 3.79 ≈ $6.14 per gallon (in Canadian dollars)
11	Make a table of each leg with distance (miles) and time. Find the total driving time.	Times ≈ 1.2 + 1.1 + 1.5 + 1.7 = 5.5 hours (Montreal leg 153 ÷ 90 = 1.7 h)

Reflection

#	Prompt	Look for
R1	When is an estimate good enough, and when do you need the exact answer?	Estimates check if answers are reasonable; exact amounts are needed for money and fuel decisions.
R2	How do you choose whether to multiply or divide with d = rt?	Multiply to find distance; divide distance by rate to find time or by time to find rate.

Answer Key – CP1

Part 1: Trip Data

#	Task	Answer
1	Find the time for each US leg (nearest hundredth of an hour) and the total US driving time in hours and minutes.	1.20 h, 1.13 h, 1.50 h → 3.83 h ≈ 3 h 50 min
2	Find the mean and median of the four leg times (Montreal leg = 153 ÷ 90 = 1.70 h).	Times 1.20, 1.13, 1.50, 1.70 → mean ≈ 1.38 h; median (1.20 + 1.50) ÷ 2 = 1.35 h
3	Estimate the total gas for the round trip (US legs twice plus the Canada leg twice, about 95 miles each way) at 31.5 mpg.	Distance ≈ 2(233.3 + 94.9) = 656.4 miles → 656.4 ÷ 31.5 ≈ 20.8 gallons
4	Is one full tank (13.2 gallons) enough to reach Montreal from Boston? Explain.	328.2 miles ÷ 31.5 ≈ 10.4 gallons < 13.2, so yes.

Part 2: Money Equations

#	Task	Answer
5	Two nights in a hotel cost $289.80. What is the price per night?	2n = 289.80 → n = $144.90
6	After buying gas, you have $472.35 left from $650. How much did you spend?	650 − x = 472.35 → x = $177.65
7	A toll is 1.25 times the regular car price of $3.60. What is the toll?	1.25 × 3.60 = $4.50
8	A snack stop costs $\frac{s}{4}$ per person, and each person paid $6.85. What was the total s?	$\frac{s}{4}$ = 6.85 → s = $27.40

Part 3: Metric Canada

#	Task	Answer
9	Convert the Burlington → Montreal leg to miles and find the time at 90 km/h in hours and minutes.	153 km ≈ 95.0 mi; 153 ÷ 90 = 1.7 h = 1 h 42 min
10	Gas in Canada is CA $1.62 per liter. Find the price per gallon in US dollars and compare with $3.39.	1.62 × 3.785 ≈ CA $6.13 ÷ 1.36 ≈ US $4.51 per gallon, about $1.12 more.
11	Build the full itinerary with leave and arrive times, distances, gas stops, and a budget that stays under $650.	Answers vary; check times from d = rt and the budget total.

Reflection

#	Prompt	Look for
R1	Why might the median leg time be a better summary than the mean?	It is less affected by one unusually long or short leg.
R2	Why must you convert units before comparing gas prices?	Liters and gallons, and US and Canadian dollars, are different units; you must compare like units.

Answer Key – Honors

Part 1: Speed & Time Analysis

#	Task	Answer
1	Find the average speed for the whole US part of the trip (total distance ÷ total time). Why is it not the mean of the three speeds?	233.3 ÷ 3.83 ≈ 61 mph; the legs take different amounts of time, so longer legs count more than shorter ones.
2	Solve d = rt for t, then find how much time is saved on the White River → Burlington leg by driving 65 mph instead of 63.4 mph (in minutes).	$\frac{95.1}{63.4} - \frac{95.1}{65}$ = 1.5 − 1.463 ≈ 0.037 h ≈ 2.2 min
3	Is speeding worth it? Use your answer and a $150 speeding ticket to argue.	No; saving about 2 minutes is not worth the risk and cost.
4	The car gets 31.5 mpg at 60 mph but 27.8 mpg at 70 mph. How much more gas does the 233.3-mile US trip use at 70 mph?	233.3 ÷ 27.8 − 233.3 ÷ 31.5 ≈ 8.39 − 7.41 ≈ 0.99 gallon

Part 2: Budget Equations

#	Task	Answer
5	Hotel tax is 0.0975 of the room price. With tax, one night is $158.43. Solve 1.0975p = 158.43 for the price.	p ≈ $144.36
6	Find the round-trip gas cost if US gas is $3.39 per gallon for 466.6 miles and Canadian gas is US $4.51 per gallon for 190 miles.	466.6 ÷ 31.5 ≈ 14.81 gal × 3.39 ≈ $50.22; 190 ÷ 31.5 ≈ 6.03 gal × 4.51 ≈ $27.20; total ≈ $77.42
7	With 2 hotel nights at $158.43 and the gas, how much is left for food and tolls? Write and solve an equation.	650 − (316.86 + 77.42) = x → x = $255.72
8	Should the family fill up in Vermont before crossing the border? Use numbers.	Yes; US gas is about $1.12 per gallon cheaper, so filling 6 gallons saves about $6.70.

Part 3: Metric Itinerary

#	Task	Answer
9	A Canadian sign says the speed limit is 100 km/h. Convert to mph.	100 × 0.621 ≈ 62.1 mph
10	Find the car's fuel efficiency in km per liter.	31.5 mi/gal × $\frac{1}{0.621}$ ÷ 3.785 ≈ 13.4 km/L
11	Create a complete itinerary (times, distances in both units, fuel stops, and costs) and justify every decision with a calculation.	Answers vary; look for correct conversions and a budget under $650.

Reflection

#	Prompt	Look for
R1	Explain why average speed for a trip is not the mean of the leg speeds.	Average speed is total distance ÷ total time; legs that take longer count more.
R2	How did writing equations help you check the budget?	Equations kept every cost organized and let you solve for the unknown amount left.